THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 3: **The World of Metals and Non-metals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The gas evolved when dilute sulphuric acid reacts with zinc is: 1

(a) sulphur dioxide **(b)** oxygen **(c)** hydrogen **(d)** carbon dioxide

2. Why do sulphur and coal burn but gold does not? 1

(a) coal is a metal **(b)** gold is less reactive
(c) sulphur is a metal **(d)** gold is gaseous

3. Which of the following pairs contains only non-metals? 1

(a) iron and carbon **(b)** carbon and sulphur
(c) gold and sulphur **(d)** copper and oxygen

4. Which metal does not react with dilute acids? 1

(a) zinc **(b)** magnesium **(c)** copper **(d)** iron

5. Why can a gold ring remain shiny for years? 1

(a) gold is soft **(b)** gold is brittle
(c) gold is unreactive **(d)** gold is a gas

6. Which of the following is the best choice for making electric wires? 1

(a) wood **(b)** sulphur **(c)** carbon (diamond) **(d)** copper

7. Which of these metals is most reactive? 1

(a) gold **(b)** copper **(c)** iron **(d)** sodium

8. Why is an iron frying pan given a wooden handle? 1

(a) wood conducts **(b)** wood is shiny
(c) iron is soft **(d)** iron conducts heat

9. Which pair is a good conductor and an insulator? 1

(a) iron and copper **(b)** rubber and plastic
(c) wood and plastic **(d)** copper and rubber

10. Which of these displaces copper from copper sulphate? 1

(a) iron **(b)** mercury **(c)** gold **(d)** silver

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does a copper vessel get a green coating over time? 2

12. Why is iodine called an exceptional non-metal? 2

13. Why is hydrogen gas called 'pop gas' and how is it tested? 2

14. Which has higher reactivity: iron or copper? How do you know? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain with reasons: 3
(a) Foil is made of aluminium
(b) Wires are made of copper
(c) Bells are made of metal.

16. A mixture of iron filings and sulphur is heated. What do you expect? 3

17. Why can a sulphur rod not be used as a wire? 3

OR

Explain how burning of coal affects the air and why its oxide is called acidic. 3

18. What are the uses of alloys and why are metals alloyed? 3

OR

Why is hydrogen not formed when acid is added to copper? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain why different metals are chosen for different uses: aluminium foil, copper wire, iron girders, gold ornaments, mercury thermometers. 5

OR

Describe the experiment to show that carbon dioxide formed by burning carbon is acidic. 5

20. Describe the displacement reactions of metals with salt solutions and how they help to arrange metals in order of reactivity. 5

OR

Compare the reaction of three metals with dilute acid and arrange them in order of reactivity. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND